Receipt Scanner Feed Rig: Construction Guide

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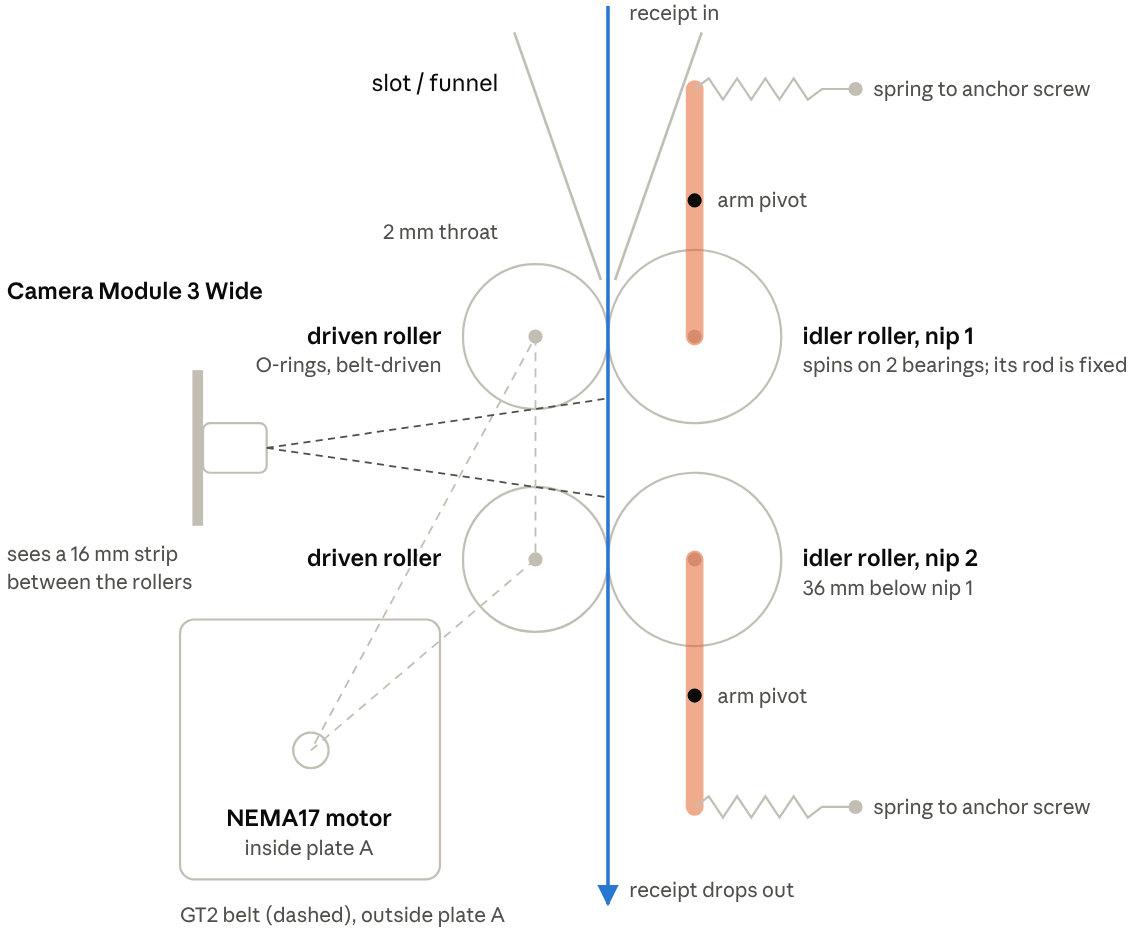
Overview

This rig answers one question before anything else gets built: can two pairs of rollers grab and move real household receipts without jamming? It also carries the camera, so the same rig later becomes the scanning prototype.

The paper path is vertical. Gravity helps the feed, and the receipt drops straight out of the bottom.

- **Funnel (slot):** you push a receipt in at the top. The 2 mm throat sits 9 mm above the first rollers, so the paper is grabbed before it can buckle.
- **Nip 1 and nip 2:** two roller pairs 36 mm apart, shorter than the shortest receipt (a 50 mm parking stub), so a receipt is always held by at least one pair.
- **Driven rollers:** O-rings grip the paper. One NEMA17 turns both through a single GT2 belt, so they always run at the same speed.
- **Idler rollers:** they press the paper against the driven rollers. Each spins on two 608 bearings around a fixed rod. Spring-loaded arms hold the rod and let the idler swing open up to 4 mm for creased paper.
- **Camera:** a Pi Camera Module 3 Wide, 55 mm from the paper, sees a 16 mm strip through the gap between the two driven rollers.
- **Frame:** two 157 × 167 mm side plates, 95 mm apart, tied by four spacers on M5 threaded rod.

The receipt drops straight down through two roller pairs



side view of the paper path, schematic, not to scale

The arms and springs sit outside the plates and the belt runs outside plate A; the side view overlaps them with the parts inside the frame.

All the source files are in `~/Documents/projects/receipt-scanner/feed-rig`:

File	What it is
`feed_rig.scad`	Every part, parametric. It prints a design report and a FIT CHECK in the console.
`stl/`	Exported parts, ready to print or cut
`bom.csv`	Parts list as a spreadsheet
`feed_test/feed_test.ino`	Stepper test sketch
`test_log.csv`	One row per receipt fed

File	What it is
`drive_detail.scad`, `idler_detail.scad`, `idler_demos/`	Visual explainers only, nothing to print

Parts list

The rig costs about $90 in parts, plus about $100 for the camera setup if you buy the Pi and camera new. Prices are rough US estimates; the same list is in `bom.csv`.

Printed (PLA or PETG, about 250 g)

Part	File	Qty	Size
Driven roller	`driven_roller.stl`	2	Ø22 × 82 mm, 4 O-ring grooves
Idler roller	`idler_roller.stl`	2	Ø28 × 82 mm, bearing pocket each end
Idler arm	`idler_arm.stl`	4	5 mm thick
Hub washer	`hub_washer.stl`	8	OD 12, ID 8.4, 6 mm thick
Pulley shim	`pulley_shim.stl`	2	OD 12, ID 8.4, 1 mm thick
Frame spacer	`spacer.stl`	4	OD 12, ID 5.5, 95 mm long
Funnel	`funnel.stl`	1	
Camera mount	`camera_mount.stl`	1	
Width shim (optional)	`width_shim.stl`	2	For 57/58 mm receipts

Cut from sheet

Part	File	Qty	Size
Side plate	`side_plate_laser.svg`	2	157 × 167 mm; this build: 4 layers of 1/8 in basswood (one per 12 × 12 in sheet), laser-cut and glued into 2 plates. Alternatives: cut from 6 mm plywood or 1/4 in MDF

Mechanical

Part	Spec	Qty	~$
Ball bearing	608ZZ, 8 × 22 × 7 mm	8	6
Smooth rod	8 mm steel, cut 2 × 129 mm + 2 × 123 mm	600 mm	10
GT2 pulley	20T, **8 mm bore**, 6 mm belt	2	6
GT2 pulley	20T, **5 mm bore**, 6 mm belt	1	3
GT2 closed belt loop	200 mm × 6 mm	1	4
O-ring	18 × 2 mm, nitrile 70A	8	8
Extension spring	about 5 mm diameter, 15–25 mm long, light	4	5
M5 threaded rod	cut 4 × 120 mm	1 m	4

Fasteners (an M2/M3/M5 assortment covers all of these, about $15)

Part	Used for	Qty
M3 × 16 socket head + nylock nut + 2 washers	Arm pivots	4 sets
M3 socket head, mixed lengths	M3 × 10: funnel 6 + camera mount 4. M3 × 12: motor 4. M3 × 16: spring anchors 4 (from the inside, nut outside)	18
M3 nut	Spring anchors	4
M3 × 6 grub screw	Driven rollers	4

Part	Used for	Qty
M3 × 4 grub screw	Idler arms	4
M2 × 6 screw	Camera board	4
M5 nut + washer	Frame spacers	8 + 8

Electronics

Part	Spec	Qty	~$
NEMA17 stepper	17HS15-1704S (already owned): 1.7 A, 5 mm D-shaft 20 **mm long**	1	12
DRV8825 driver	STEP/DIR module with heatsink (ordered)	1	6
Capacitor	100 µF, 25 V or higher	1	1
Arduino Nano/Uno or ESP32	runs `feed_test.ino`	1	8
Power supply	12 V 2 A + DC jack	1	10
Jumper wires + breadboard		1	6
Pi Camera Module 3 **Wide**		1	35
Raspberry Pi 4 (2 GB or more)	Pi 5 needs a 22-to-15-pin camera cable	1	45
Camera ribbon cable	15-pin, 300 mm	1	4
microSD card + Pi power supply	32 GB; 5 V 3 A USB-C	1 each	18

Lighting is not designed yet.

Tools and before you start

Measure your sheet material first: every shaft length and pulley position depends on the plate thickness.

Tools: Snapmaker (printer + laser or CNC), calipers, hacksaw or rotary tool with a cut-off disc, metal file, M3 tap, 8 mm drill bit or reamer, 3.5 mm and 5.5 mm drill bits, hex keys (1.5, 2, 2.5 mm), small screwdriver, CA glue, wood glue (only if laminating 3 mm sheets).

1. **Measure the plate material** with calipers in three places. "1/4 in" or "6 mm" sheet is often 5.2–6.4 mm.
2. **Open `feed_rig.scad`** in OpenSCAD and set `plate_t` to the average you measured.
3. **Set the cutting mode:** `plate_mode = "laser"` for the laser (the CNC uses its own side_plate_cnc_inner / side_plate_cnc_outline exports and ignores `laser_kerf`), or `"print"` to print the plates.
4. **Press F5 and read the console.** Check that every line of the FIT CHECK says `ok`. If one says `FAIL`, the line names the clash and its size in mm.
5. **Note the report values** you will use later: driven shaft length, idler rod length, belt loop length and steps/mm.
6. **Export the parts.** OpenSCAD is not on your terminal path, so use the full path, from the project folder:

```
cd ~/Documents/projects/receipt-scanner/feed-rig && for p in driven_roller
idler_roller idler_arm hub_washer pulley_shim spacer funnel width_shim
camera_mount; do /Applications/OpenSCAD-2021.01.app/Contents/MacOS/OpenSCAD -
D "part=\"$p\"" -o stl/$p.stl feed_rig.scad; done
```

```
/Applications/OpenSCAD-2021.01.app/Contents/MacOS/OpenSCAD -D
'part="side_plate_2d"' -o stl/side_plate_laser.svg feed_rig.scad
```

For Snapmaker Luban, use the `*_luban.svg` files (for example `stl/side_plate_laser_luban.svg`). They hold only black cut lines with no fill, at 1 unit = 1 mm. To regenerate every cut file after a change, run `tools/export_cut_files.sh`.

The files already in `stl/` match the default settings (6 mm plates, laser). Re-export only if you change a parameter.

Fabrication

Make one test piece of each critical fit before cutting or printing the full set. It costs minutes and saves a second sheet.

1. Test fits first

- [ ] **Bearing in plate:** cut a small square with one bearing hole. A 608 should press in by hand or with a vise, not drop through. `Too loose: raise bearing_press by 0.05 (the hole gets smaller). Too tight to press in: lower bearing_press by 0.05`.
- [ ] **Bearing in idler roller:** print 10 mm of the idler roller end. A 608 should press in firmly. Adjust the `+ 0.15` in `idler_roller()`.
- [ ] **Roller bore on rod:** the driven roller should slide onto the 8 mm rod snugly, with no wobble. Ream with an 8 mm bit if tight.
- [ ] **Arm pivot:** an M3 bolt should turn freely in the arm's 3.5 mm pivot hole.

2. Side plates (2)

- **Laser:** use basswood or poplar plywood, not hardware-store birch. Expect 2–3 passes for 6 mm with air assist on. Cut both plates in the same orientation.
- **CNC (for MDF or hardware-store plywood):** saw the panel into blanks of about 185 × 195 mm first. Use one 1/8 in (3.175 mm) flat end mill for everything, about 0.5–1 mm per pass, 300–500 mm/min. Import side_plate_cnc_inner_luban.svg and side_plate_cnc_outline_luban.svg at the same X/Y (they share a bounding box), set both to Luban's On the Path toolpath, cut the inner file first, then the outline with tabs. The paths are already offset for the bit, so every hole comes out at its true size, including the 3.4 mm M3 holes.
- **MDF:** coat the inside of each bearing hole with a thin layer of CA glue and let it harden before pressing bearings in.
- **Printed plates:** export with `plate_mode="print"`. Use 4 walls and 25–30% infill.

3. Printed parts

Part	Qty	Orientation	Settings
Driven roller	2	Upright, as exported	5 perimeters around the set-screw holes, slow outer wall (about 25 mm/s)
Idler roller	2	Upright	4 perimeters, slow outer wall
Idler arm	4	Flat	100% infill

Part	Qty	Orientation	Settings
Funnel	1	On its end cap, as exported	The top cap bridges about 30 mm; add support if your bridging is weak
Camera mount	1	Bar face down, as exported	Standoffs up
Hub washer, pulley shim, spacer, width shim	8, 2, 4, 2	As exported	Any

Common settings: 0.16–0.2 mm layers, 30–40% infill, elephant's foot compensation on. PLA is fine for this rig. Keep it out of hot cars and sunny windows.

4. Metal parts

Part	Qty	Length	Notes
Driven shaft (8 mm rod)	2	129 mm	File a small flat where each set screw lands: 4 mm from each roller end and at the pulley
Idler rod (8 mm rod)	2	123 mm	File a small flat where each arm's set screw lands
M5 threaded rod	4	120 mm	File the cut ends so nuts thread on

Deburr and chamfer every rod end so it slides through bearings without scraping.

5. Prepare the printed parts

- [ ] Ream the driven roller bores with an 8 mm bit by hand.
- [ ] Tap the roller set-screw holes (2.7 mm) with the M3 tap, or thread a screw in once.
- [ ] Drill or ream each arm's rod hole to a snug fit on the 8 mm rod.
- [ ] Clean the M5 spacer bores with a 5.5 mm bit if the rod binds.

Assembly

Build everything onto plate B lying flat, then close the frame with plate A on top. Plate A is the motor side: the motor, pulleys and belt all go on it.

The idler rod never turns. The idler roller spins around it on two bearings, and the rod only swings a few millimetres with the arms. The `idler_demos/` files animate this.

A. Bearings and rollers

1. **Press a 608 into each round bearing hole** in both plates (4 total). Make each bearing flush with the plate's *inner* face. A 6 mm plate leaves about 1 mm sticking out on the outside; that is expected.
2. **Fit the O-rings:** roll 4 O-rings into the grooves of each driven roller.
3. **Mount each driven roller on a 129 mm shaft.** Center it so the long end sticks out about 32 mm on one side (the pulley side) and the short end about 15 mm on the other. Tighten its two M3 × 6 set screws onto the filed flats, snug only.
4. **Press a 608 into each end of both idler rollers** (4 total), flush with the roller ends.

B. Build on plate B

5. **Lay plate B flat, inner face up.** It is the plate without the pulleys, so either cut plate works.
6. **Driven rollers:** slide a hub washer onto the short end of each driven shaft. Push that end into a plate B bearing, so the washer sits between the roller and the bearing's inner ring.
7. **Idler rollers:** push an idler rod through each arc slot in plate B from the outside, leaving about 8 mm outside for the arm. Thread on a hub washer, the idler roller, then a second hub washer.
8. **Funnel:** stand it on plate B over its 3 holes, wide mouth at the top edge. Fix it with three M3 × 10 screws from the outside of plate B.
9. **Spacers:** push an M5 rod through each spacer hole in plate B with a washer and nut outside, and slide a 95 mm spacer onto each rod.
10. **Add the second hub washers** on the long ends of the driven shafts.

C. Close the frame

11. **Lower plate A on top.** Guide in order: both driven shafts into the plate A bearings, both idler rods through the arc slots, then the four M5 rods.
12. **Fasten:** M5 washer and nut on each rod. Tighten the four nuts a little at a time in a diagonal pattern, like a car wheel. Then fix the funnel to plate A with three M3 × 10 screws.

13. **Check:** both driven shafts should spin freely by hand. If they bind, the frame is twisted: loosen the M5 nuts, square the plates, and retighten.

D. Arms and springs

14. **Fit the arms** on each idler rod outside both plates (4 arms). Bolt each arm's pivot hole to the matching plate hole with an M3 × 16, a washer each side, and a nylock nut. Tighten until there is no wobble but the arm still swings freely.
15. **Lock the rod:** tighten each arm's M3 × 4 set screw onto the rod flat. Both arms on one rod must sit at the same angle, so the rod stays parallel to the driven roller.
16. **Spring anchors:** put an M3 × 16 through the second anchor hole of each row from the inside, and tighten a nut on the outside face. About 7 mm of shank sticks out for the spring to hook onto. This is the default nip force; holes further out press harder.
17. **Springs:** hook each spring from the hole at the top of its arm (the tail) to its anchor screw. The lower arms point down, so their tails are at the bottom.
18. **Check:** each idler should press against its driven roller. Push it away by hand: it should open about 4 mm and snap back.

E. Motor and belt

19. **Motor:** hold the NEMA17 against the *inside* face of plate A, with its shaft through the large slotted hole. Fix it with four M3 × 12 through the slots, loosely (M3 × 10 is too short with laminated 6.35 mm plates).
20. **Pulleys:** slide a 1 mm pulley shim, then an 8 mm-bore pulley, onto the long end of each driven shaft. Put the 5 mm-bore pulley on the motor shaft. Mount all three pulleys hub first, so the set-screw hub faces the plate and the toothed part faces out. The motor's 20 mm shaft is only long enough for the hub this way; its tip ends about 3 mm inside the toothed part, which is fine. Set the motor pulley's screw on the shaft's flat. Line up all three in one plane with a straightedge, then tighten the set screws onto the flats.
21. **Belt:** loop the 200 mm belt around all three pulleys. Slide the motor along its slots until the belt is firm but not tight: it should twist about 90° by hand at the middle of its longest run. Tighten the four motor screws.
22. **Check:** turning one driven shaft by hand turns the other and the motor.

F. Camera (after test 1)

23. **Plug in the ribbon cable** to the Camera Module 3 Wide first, then fix the board to the mount's standoffs with four M2 × 6 screws, lens facing out, connector at the bottom.

24. **Mount it:** slide the mount between the plates on the driven-roller side, behind the rollers. Fix it with four M3 × 10 through the camera slots into the bar ends.
25. **Set the distance:** slide it along the slots until the lens front is about 55 mm from the paper plane, then tighten. Autofocus covers small errors.
26. **Route the ribbon:** down from the connector, then bent back under the mount bar and out of the frame. The motor sits directly below, so keep the ribbon clear of it.

Electronics and first run

The stepper runs from a DRV8825 driver, controlled by a Raspberry Pi Pico running pico/stepper_test.py (or an Arduino running `feed_test.ino` over USB serial; the steps below use the Arduino sketch, using Arduino pins D2–D4 and its 5 V pin in place of the Pico's GP2–GP4 and 3V3). Wire everything with the 12 V supply unplugged.

Wiring

DRV8825 pin	Connects to	Note
STEP	Pico GP2 (pin 4)	
DIR	Pico GP3 (pin 5)	
EN	Pico GP4 (pin 6)	Active low; the script pulls it low to run
RESET and SLEEP	Pico 3V3 OUT (pin 36)	Both must be high, or the driver stays off
M2	Pico 3V3 OUT	With M0 and M1 low: 1/16 microstepping
M0, M1	Pico GND	
GND (logic side)	Pico GND (pin 38)	Shares ground with the 12 V supply
VMOT	12 V supply +	100 µF capacitor across VMOT and GND, close to the driver, stripe to GND
GND (motor side)	12 V supply −	
A1, A2 / B1, B2	Motor coil A / coil B	Find the pairs with a multimeter: a few ohms between the two wires of one coil

Driver current: on most DRV8825 boards the current is 2 × Vref. Measure Vref between the trimmer and GND with the 12 V on, and set it to 0.45–0.5 V for 0.9–1.0 A; the 17HS15-1704S is rated 1.7 A and this rig needs little torque. Stick the heatsink on the driver chip. The DRV8825 whines more than a TMC2209; that is normal.

Never plug or unplug the motor while the 12 V supply is on. It can destroy the driver.

Flash and run

1. Open `feed_test/feed_test.ino`. Check that `ROLLER_R_MM` matches "Effective roller radius: driven" in the OpenSCAD report (11.6 with the defaults).
2. Upload it, open the serial monitor at 115200 baud, then switch on the 12 V supply.
3. Send `v 30`, then `r`. The rollers should turn smoothly. Send `x` to stop.
4. **Direction:** the paper must travel *down*. If the nip surfaces move up, send `d` to reverse, or swap one coil's two wires.
5. **Speed check:** send `f 100`. A pencil mark on the O-ring should travel about 100 mm (about 1.4 turns of the roller).

Command	Action
`v <mm/s>`	Set paper speed (default 60)
`r`	Run continuously
`f <mm>`	Feed a fixed distance, then stop
`x`	Stop
`d`	Reverse direction, to back out a jam

Camera check (Raspberry Pi)

With the ribbon in the Pi's camera connector and Raspberry Pi OS installed, run `rpicam-hello --list-cameras`. It should list an IMX708 wide sensor. Then `rpicam-hello` shows a live preview: hold a receipt in the nip and check that the text is sharp.

Testing and tuning

Test 1 passes when the rig feeds fewer than 1 in 50 flat, curled and creased receipts wrongly, and handles both short and long ones. Crumpled receipts are logged for information only.

Receipt set (50 real receipts)

Category	Count	Notes
Flat	15	The baseline
Curled	10	Fresh off the roll, not flattened
Creased	10	Folded once or twice, from a wallet or pocket
Crumpled	5	Lightly crumpled, then smoothed by hand
Short	5	Under 80 mm
Long	5	Over 300 mm

Number each receipt (R001…R050) so you can repeat runs with the same set.

Protocol

1. Run the motor continuously at the test speed (start at 60 mm/s).
2. Insert a receipt into the slot, print side toward the camera side, until nip 1 grabs it. Let go.
3. Log one result per receipt in `test_log.csv`: `ok`, `skew` (with an estimated angle), `stall`, `jam`, `no_grab` or `double`.
4. Change one setting at a time and note it in the `config_note` column.
5. Sweep the three main variables: nip force (anchor holes 1–4), speed (30, 60, 100 mm/s) and driven surface (O-rings or silicone tube).

Tuning

Symptom	Try
`no_grab`	More nip force (anchor further out); lower `throat_h`; grippier driven surface (`tube`)

Symptom	Try
Buckling above nip 1	Smaller `throat_gap` (1.5 mm); lower `throat_h`; slower speed
`skew`	Width shims for narrow receipts; equal spring force on both arms; more O-rings
Short receipts stall	Shorter `nip_pitch` (the FIT CHECK confirms the rollers still fit)
Curl catches the funnel lip	Wider `funnel_mouth`
Creases jam	More `idler_travel`; softer springs with more speed
Uneven speed or slip	Ignore unless the paper stops: the camera software will measure real paper motion

Reference

All values below are the `feed_rig.scad` defaults, with 6 mm plates cut on the laser.

Key dimensions

Item	Value
Plate size	157 × 167 mm, 6 mm thick
Plate inner gap	95 mm
Roller length	82 mm
Driven roller grip diameter	23.2 mm (22 mm core + O-rings 0.6 mm proud)
Idler roller diameter	28 mm
Nip 1 to nip 2	36 mm
Funnel throat	2 mm wide, 9 mm above nip 1
Idler travel	4 mm open, 1 mm overtravel
Nip force per arm	spring force × 0.82

Item	Value
Belt loop	200 mm (motor slot covers 196–204 mm)
Steps per mm	43.9 at 1/16 microstepping (DRV8825)
Camera	55 mm from paper; 136 mm wide field; 16 mm visible strip
Camera resolution	287 dpi at 1536 × 864, 120 fps; 431 dpi at 2304 × 1296, 56 fps

Holes as drawn in the plate file (laser mode)

Hole	Drawn	Cuts to about	For
Bearing	21.75 mm	21.9 mm	608 press fit, 0.1 mm interference
M3	3.25 mm	3.4 mm	Clearance
M5	5.35 mm	5.5 mm	Clearance
Motor boss	22.85 mm, slotted ±3 mm	23 mm	NEMA17 boss (22 mm)
Idler rod slot	8.85 mm wide arc	9 mm	8 mm rod, 4 mm travel

Fit check (verified on the current design)

Check	Clearance
Funnel screw heads vs nip 1 arm, closed and open	2.2 mm
Spring anchor screw heads vs arms	9.4 mm
Wood between bearing hole and idler slot	9.1 mm
Funnel cap material above its screw holes	1.65 mm

Check	Clearance
Motor pulley hub on the motor shaft	5 mm to spare for the hub, teeth overhang 3 mm (20 mm shaft, hubs toward the plate)
Camera mount above motor body	20 mm
Roller end play between hub washers	1 mm
Gap between the two idler rollers	8 mm

Troubleshooting

Problem	Likely cause	Fix
Driven shafts bind after closing the frame	Plates twisted or not parallel	Loosen the M5 nuts, square the frame, retighten diagonally
Bearing loose in the plate	Hole cut too big	CA glue around it, or recut with a higher `bearing_press`
Arm doesn't swing freely	Pivot nut too tight, or pivot hole too small	Back off the nylock; drill the arm to 3.5 mm
Idler doesn't touch the driven roller	Arms locked at different angles	Loosen one arm's set screw, let the spring seat the rod, retighten
Belt jumps teeth	Belt too loose or pulleys out of line	Retension; realign the pulleys with a straightedge
Motor buzzes but doesn't turn	Coil pair swapped, or current too low	Recheck the coil pairs; raise Vref a little
Rollers slip on paper	Too little nip force	Move the springs to an outer anchor hole
Camera image blurry	Distance or motion blur	Check 55 mm lens-to-paper; shorten exposure or slow the paper